1. Create Database

Purpose

Creates a new database to store supermarket data.

Query:

CREATE DATABASE supermarket_db;

---

2. Select Database

Purpose

Selects the database for performing SQL operations.

Query:

USE supermarket_db;

---

3. Create Table

Purpose

Creates a table to store product information.

Query:

```sql
CREATE TABLE stock_management (
  p_id INT,
  p_name VARCHAR(30),
  p_price INT,
  p_stock INT
);
```

4. INSERT INTO

Purpose

Adds records to the table.

Query:

INSERT INTO stock_management VALUES (110,'paneer',25,20);

Other inserted products:

- Paneer
- Milk
- Curd

- Pen
- Paper
- Pencil
- Tissue

Note: Product ID **110** is inserted twice, so duplicate records exist because no PRIMARY KEY constraint was defined.

5. SELECT Statement

Purpose

Displays all records.

Query:

SELECT * FROM stock_management;

Result:

p_id	p_name	p_price	p_stock	p_brand
110	paneer	70	20	NULL
110	paneer	70	20	NULL
111	milk	35	21	NULL
112	curd	50	22	NULL
113	pen	55	20	NULL
114	paper	70	20	NULL
115	pencil	90	20	NULL
116	tissue	10	19	NULL

---

6. ALTER TABLE

Purpose

Adds a new column to the existing table.

Query:

ALTER TABLE stock_management ADD p_brand VARCHAR(20);

---

7. UPDATE Statement

Purpose

Modifies existing data.

Query:

UPDATE stock_management SET p_price = 70 WHERE p_id = 110;

Result

---

8. SQL_SAFE_UPDATES

Purpose

Disables MySQL Safe Update mode.

Query:

SET SQL_SAFE_UPDATES = 0;

Why?

Allows UPDATE and DELETE statements even when MySQL considers them unsafe.

---

9. WHERE Clause

Purpose

Filters rows based on a condition.

Query:

SELECT p_id, p_name FROM stock_management WHERE p_price > 50;

Result

Result Grid | Filter Rows:

p_id	p_name
110	paneer
110	paneer
114	paper
113	pen
115	pencil

---

10. ORDER BY

Purpose

Sorts records.

<u>Ascending Order</u>

SELECT p_id, p_name FROM stock_management WHERE p_price > 50 ORDER BY p_name;

Result:

Result Grid | Filter Rows:

p_name	total_stock
paneer	2
milk	1
curd	1
pen	1
paper	1
pencil	1
tissue	1

Sorts product names alphabetically.

<u>Descending Order</u>

SELECT p_id, p_name FROM stock_management ORDER BY p_id DESC;

Displays products from highest Product ID to lowest.

Result:

Result Grid | Filter Rows:

p_id	p_name
116	tissue
115	pencil
114	paper
113	pen
112	curd
111	milk
110	paneer
110	paneer

---

11. GROUP BY with COUNT()

Purpose

Groups identical product names and counts the number of records.

Query:

SELECT p_name, COUNT(p_price) AS total_stock FROM stock_management GROUP BY p_name;

---

12. DELETE Statement

Purpose

Removes records from a table.

Query:

DELETE FROM stock_management WHERE p_id = 109;

13. GROUP BY with SUM()

Purpose

Calculates the total stock available for each product.

Query:

SELECT p_name, SUM(p_stock) AS total_stock FROM stock_management GROUP BY p_name;

Result:

Result Grid | Filter Rows:

p_name	total_stock
paneer	40
milk	21
curd	22
pen	20
paper	20
pencil	20
tissue	19